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Paleomagnetic considerations of the development of the Pennsylvania salient in the central Appalachians

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Abstract

Late Paleozoic remagnetizations of central Appalachian clastic and carbonate rocks of Ordovician to Devonian age constrain the structural evolution of the Pennsylvania salient. New paleomagnetic results from 26 sites located on nine folds combined with previously published results from 60 sites yield a mean direction for the northeastern limb of the salient at 172°/1° ($k = 49$) and a mean direction for sites southwest of the salient at 173°/1° ($k = 70$). These two directions are not statistically different from each other and are not significantly rotated from the expected Permian paleomagnetic direction for North America. Individual fold test results show that this secondary magnetization is synchronous with or even older than folding across the region. These paleomagnetic results suggest that curvature of fold axes around the salient is an original feature of fold growth and not the result of post-orogenic oroclinal bending.

In contrast, analysis of the primary magnetization in the Silurian Bloomsburg Formation shows that directions from the two limbs of the salient are significantly different. New results from twenty sites yield a mean direction for the northeastern limb at 4°/−40° ($k = 31$), and for the southwestern limb at 338°/−38° ($k = 23$). Combining these results with previously published directions from the Bloomsburg Formation yields a mean direction for the northeastern limb at 1°/−34° ($k = 32$) and for the southeastern limb at 343°/34° ($k = 30$). The difference in declination does not appear to be the result of internal deformation of the rocks or acquisition of magnetization during a period of rapid apparent polar wander. Rather, these directions indicate that the northeastern limb has been rotated clockwise 18 ± 9° relative to the southwestern limb. The sense and amount of this rotation is consistent with the curvature of the orogen around the salient. However, within each limb of the salient the primary component declinations do not directly correlate with the orientation of the first-order fold axes. Collectively therefore, these paleomagnetic results suggest that the structural evolution of the salient included pre-orogenic rotations or rotations that occurred early in the Alleghanian orogeny followed by later remagnetization and folding of the central Appalachian foreland.

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1. Introduction

The nature and origin of the salients and recesses in the Appalachian mountains of North America have been debated since the early work of Rogers (1858). These arcuate features are defined by curvature of the structural grain along the axis of the mountain belt. Two end-member models have been proposed to explain their origin: (1) The arcs are primary features that reflect sediment accumulation and subsequent translation over an irregular basement margin (Rankin, 1976; Thomas, 1977). In this model, fold axes developed with their present curved trends. (2) The arcs are secondary features in which the fold axes were originally more linear but were rotated or bent into their present orientation by subsequent tectonic deformation. This second alternative has traditionally been termed oroclinal bending with the resulting arcs defined as oroclines (Carey, 1955). Marshak (1988) has termed these two alternatives as nonrotational and rotational arcs respectively.

Paleomagnetic analyses are often used to test between the two models. In the simplest test, paleomagnetic declinations from coeval rocks around the arc are compared with each other or against a reference direction derived from a reference apparent polar wander (APW) curve. If the declination differences are significant and of the correct sense, then a rotational origin for the arc is supposed. More rigorous tests utilize linear regression techniques, either by correlating bedding strike against declination (Schwartz and Van der Voo, 1983) or rotation (declination minus an appropriate reference declination) against the trend of the fold axes (Eldredge et al., 1985; Lowrie and Hirt, 1986).

The orientation of folds around the Pennsylvania salient changes nearly 60° (Fig. 1). Fold axes on the northeastern limb of the salient trend nearly east–west while those on the southwestern limb of the salient trend north 30° east. Early paleomagnetic investigations in the region produced contradictory results. Differences between the paleomagnetic declinations of the Upper Silurian Bloomsburg Formation (Irving and Opdyke, 1965; Roy et al., 1967) and Middle Silurian Rose Hill Formation (Graham, 1949) suggested at least partial oroclinal bending while the similarity of directions from the Lower Carboniferous Mauch Chunk Formation around the salient (Knowles

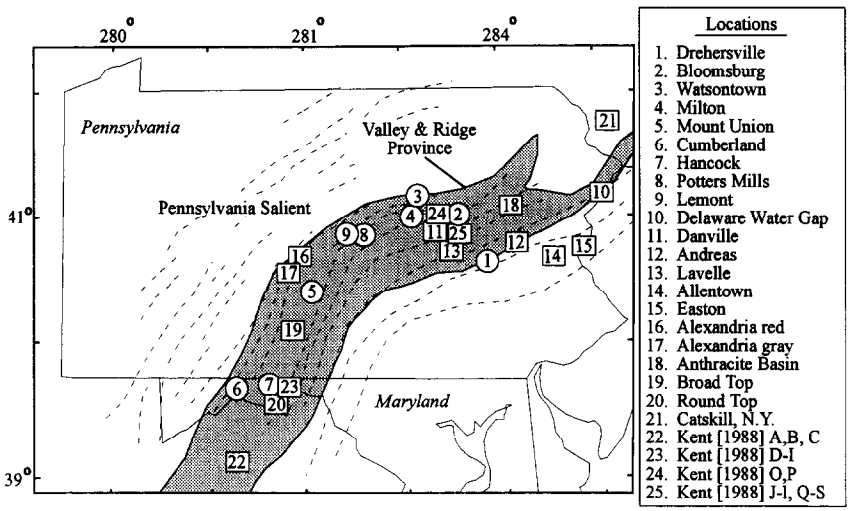


Fig. 1. Map of Pennsylvania showing the fold localities for this analysis (circles) as well as localities from previous paleomagnetic investigations (squares) used in this study. Dashed lines show the trends of the major folds around the Pennsylvania salient. Shaded region is the Valley and Ridge Province.

and Opdyke, 1968) supported a nonrotational origin. Using data from four Paleozoic redbed units (the Ordovician Juniata Formation, Silurian Rose Hill and Bloomsburg formations and the Devonian Catskill Formation) and the more rigorous linear regression tests, Schwartz and Van der Voo (1983) and Eldredge et al. (1985) also concluded that the salient was not a secondary feature.

More recent paleomagnetic work has shown that many of the originally reported directions from these redbeds, especially from sites on the southwestern limb of the salient are contaminated by secondary magnetizations (e.g. Kent and Opdyke, 1985). Using detailed thermal demagnetization techniques, Kent and Opdyke (1985) were able to isolate two ancient magnetizations in the redbeds, a characteristic magnetization and a secondary chemical remagnetization acquired during Alleghanian deformation in the Permian. Based on declination differences of the characteristic components in the Bloomsburg (Kent, 1988), Catskill (Miller and Kent, 1986a,b) and Mauch Chunk formations (Kent and Opdyke, 1985; Kent, 1988), it was concluded that oroclinal bending accounts for approximately half of the change in structural trend around the salient (Kent, 1988). The absolute sense of rotation from this analysis could not be determined because of the scarcity of reliable Silurian, Devonian and Carboniferous reference poles for North America.

The significance of the late Paleozoic remagnetization as an indicator of tectonic rotations has not been fully explored in these previous paleomagnetic investigations. Yet, these secondary, remagnetized directions also provide important information about the tectonic development of the salient, especially because remagnetization coincides with the late Paleozoic tectonism that deformed the Appalachian foreland. In this paper, the development of the Pennsylvania salient as a primary or secondary structure relative both to the characteristic magnetizations of the Bloomsburg and Mauch Chunk formations and the late Paleozoic remagnetization is investigated. Paleomagnetic results from 26 new sampling sites from nine fold localities are combined with previously reported results around the salient in Pennsylvania, New York and Maryland. These results provide new constraints on the structural development of the Pennsylvania salient.

2. Regional setting

The central Appalachian orogenic belt consists of a nearly complete sequence of Paleozoic sedimentary rocks deposited on a Grenville-age metasedimentary basement. The orogen has experienced at least three phases of Paleozoic compressional tectonics, the Taconic Orogeny in the Ordovician, the Acadian Orogeny in the Devonian, and the Alleghanian Orogeny in the Late Carboniferous or Permian. Development of the foreland fold and thrust structures in the Valley and Ridge portion of the orogen is primarily attributed to the Alleghanian orogeny (Faill, 1985; Hatcher et al., 1989), when North America collided with the northwest margin of Africa.

The regional structure of the central Appalachian foreland consists of a cover sequence of Ordovician through Pennsylvanian clastic rocks folded above a duplex of Cambrian to Ordovician carbonates (Gwinn, 1964; Kulander and Dean, 1986; Geiser, 1989). Structural analyses in the Pennsylvania anthracite region have shown that folding is part of a nearly continuous deformation sequence that also includes stages of prefolding, layer-parallel shortening and top-to-the-foreland, bedding-parallel shear (Nickelsen, 1979; Gray and Mitra, 1993). Fold growth in the roof layer is linked to displacements of the individual horses of the underlying duplex (Ferrill and Dunne, 1989; Meyer and Dunne, 1990), although some folds are also related to the propagation of thrust faults from the duplex into the overlying cover rocks (Gray, 1991).

In the central Appalachians, the clastic redbeds have provided the more reliable paleomagnetic data. Primary magnetizations have been reported for the Ordovician Juniata Formation (Van der Voo and French, 1977; Miller and Kent, 1989), the Middle Silurian Rose Hill Formation (French and Van der Voo, 1979), the Silurian–Devonian Andreas redbeds (Miller and Kent, 1988a), the Upper Devonian Catskill Formation (Miller and

Kent, 1986a,b), and the Lower Carboniferous Mauch Chunk Formation (Knowles and Opdyke, 1968; Kent and Opdyke, 1985; Kent, 1988). In these rocks the primary magnetization resides in hematite with discrete unblocking temperatures above 650°C. These primary magnetizations pass fold tests and often include dual polarity directions. The primary magnetization in the Mauch Chunk Formation has been used to document the polarity stratigraphy for Visean to Namurian times (DiVenere and Opdyke, 1991).

The age of characteristic magnetization of the Upper Silurian Bloomsburg Formation was suspect because incremental fold tests of this component showed a significant synfolding distribution (Kent, 1988). However, more recent analysis indicates that the characteristic directions in these rocks may have been rotated at the grain scale by penetrative strains associated with Alleghanian deformation (Stamatakos and Kodama, 1991a). By considering only those sites with relatively low strain values, Stamatakos and Kodama (1991a) were able to obtain a prefolding Silurian magnetization for these rocks. The characteristic magnetization of the Silurian–Devonian Andreas redbeds also showed a significant synfolding distribution (Miller and Kent, 1988a). Yet, because the resulting pole position is similar to other earliest Devonian paleomagnetic pole positions, Miller and Kent (1988a) argued that the age of this magnetization is Devonian. They interpreted the fold test results as either a strain effect or as evidence for minor Devonian folding in the central Appalachians.

Table 1
Paleomagnetic sampling sites

Fold location	Site	Fm.	Lat	Long	DD	Dip
Drehersville, PA	SbA	Sbl	40.6	284.0	312	58
Drehersville, PA	SbB	Sbl	40.6	284.0	320	63
Drehersville, PA	SbC	Sbl	40.6	284.0	220	47
Drehersville, PA	SbD	Sbl	40.6	284.0	208	54
Bloomsburg, PA	SbE	Sbl	41.0	283.6	146	28
Bloomsburg, PA	SbF	Sbl	41.0	283.5	335	32
Watsontown, PA	SbG	Sbl	41.1	283.2	16	42
Watsontown, PA	SbH	Sbl	41.1	283.2	179	15
Watsontown, PA	SbI	Sbl	41.1	283.2	15	12
Watsontown, PA	SbJ	Sbl	41.1	283.2	8	25
Watsontown, PA	SbK	Sbl	41.1	283.2	354	14
Watsontown, PA	SbL	Sbl	41.1	283.2	15	5
Milton, PA	SbM	Sbl	41.0	283.2	7	77
Milton, PA	SbN	Sbl	41.0	283.1	184	55
Mount Union, PA	SbO	Sbl	40.4	282.1	112	28
Mount Union, PA	SbP	Sbl	40.4	282.2	310	28
Cumberland, MD	SbQ	Sbl	39.7	281.3	276	47
Cumberland, MD	SbR	Sbl	39.7	281.3	120	20
Hancock, MD	SbS	Sbl	39.7	281.6	295	66
Hancock, MD	SbT	Sbl	39.7	281.6	295	77
Hancock, MD	SbU	Sbl	39.7	281.6	130	60
Hancock, MD	SbV	Sbl	39.7	281.6	130	80
Andreas, PA	SDaA	SDarb	40.8	284.2	332	42
Andreas, PA	SDaB	SDarb	40.8	284.2	155	39
Potters Mills, PA	OrA	Or	40.8	282.3	153	36
Potters Mills, PA	OrB	Or	40.8	282.3	331	47
Lemont, PA	ObA	Obd	40.9	282.3	342	48
Lemont, PA	ObB	Obd	40.9	282.3	97	20

Fm. is the formation sampled: Sbl = Silurian Bloomsburg Formation; SDarb = Silurian–Devonian Andreas redbeds; Or = Ordovician Reedsville Formation; Obd = Ordovician Bellefonte dolomite. Lat and Long are the latitude (°N) and longitude (°E) of the sampling site, respectively. DD and Dip are the dip direction (°) and dip (°) of bedding.

Ancient magnetizations have also been reported in the Ordovician Martinsburg Formation (Hower, 1979), the Silurian–Ordovician Beemerville Complex (Proko and Hargraves, 1973), the Upper Carboniferous Brush Creek limestone and Buffalo siltstone (Payne et al., 1981), and the Upper Carboniferous or lowermost Permian Dunkard Formation (Helsley, 1965). However, some or all of these magnetizations may not be primary because none of the results yielded positive fold tests. The Beemerville magnetization is carried by secondary hematite and its age is not constrained by any paleomagnetic field tests. The tectonic corrections used in the Martinsburg study are complex and interpreted from sparse structural data. Moreover, these tectonic corrections did not significantly improve the large scatter of the data. The Buffalo siltstone directions appear to be contaminated by recent overprints (Van der Voo, 1990). The Bush Creek limestone directions may not completely average secular variation because of the rocks limited exposure (Payne et al., 1981).

Late Paleozoic remagnetized directions have been uncovered in many of the carbonates and in nearly all of the clastic redbeds throughout the central Appalachians (see McCabe and Elmore, 1989 for a comprehensive review of the late Paleozoic remagnetization in the Appalachians). In the limestones and shales the secondary magnetization usually resides in magnetite. In the redbeds, this secondary magnetization is often carried by fine-grained hematite. Although the exact nature of the remagnetization process is not yet known, growth of new magnetic phases as the rocks interacted with chemically active brines during the Alleghanian orogeny is considered the most probable remagnetization mechanism (Oliver, 1986).

On a regional scale, the acquisition of the remagnetization coincides with folding (Kent and Opdyke, 1985). However, more detailed examination shows that the timing of remagnetization relative to folding may vary across the fold and thrust belt. Local fold tests of the Bloomsburg Formation show that the secondary magnetization is postfolding at the Delaware Water Gap folds near the hinterland portion of the belt but synfolding at Montour Ridge in the central portion of the belt (Stamatakos and Kodama, 1991a). Perry (1991) showed that the remagnetization of the Silurian Clinton ironstone in the foreland portion of the belt was prefolding. This variation of local fold-test results suggests relatively rapid remagnetization across the region as folding propagated more slowly toward the foreland.

3. Sampling and laboratory procedures

Because of the variation in fold test results, local fold tests are important to constrain the geometry of bedding at the time of remagnetization. Samples were collected from sites on both limbs of folds across and along the strike of the foreland fold and thrust belt in Pennsylvania and Maryland (Table 1; Fig. 1). Four rock units were sampled, the Ordovician Reedsville shale and Bellefonte dolomite, the Silurian Bloomsburg Formation and the Silurian–Devonian Andreas redbeds. The new Andreas sites were collected on freshly exposed outcrops in the same roadworks quarry sampled by Miller and Kent (1988a), and were used to augment their results. Samples were drilled either in the field using a portable drill fitted with a 2.5 cm diameter diamond coring bit or from hand samples using a similar-sized coring device. Samples were oriented using a standard paleomagnetic orientation device and magnetic compass. Bedding orientations were measured in the field with a Brunton compass. The magnetization of these rocks is too weak to deflect the compass readings.

Progressive thermal demagnetization (8–15 steps) was performed on all samples using a Schonstedt TSD-1 thermal demagnetizer. Remanence measurements were made on a three-axis cryogenic magnetometer. Magnetic components were determined by fitting a least square regression line to the linear portions of the demagnetization trajectories. The distribution of sample directions was calculated using Fisher statistics (Fisher, 1953). Incremental fold tests were performed using 5% increments of limb dip unfolded about the line of strike. The McFadden and Jones (1981) method was used to determine the statisti-

cal significance of each fold test. Demarest's (1983) method was used to evaluate the statistical uncertainties associated with comparisons of paleomagnetic declinations.

4. Results

Thermal demagnetization of the Bloomsburg and Andreas samples uncovered three magnetic components similar to thermal demagnetization results of many other Appalachian redbeds: (1) an initial component parallel to the direction of the present earth's field with unblocking temperatures less than 400°C; (2) a secondary or B component with distributed unblocking temperatures between 300 and 660°C; and (3) a characteristic or C component with discrete unblocking temperatures between 660 and 680°C (Fig. 2a and

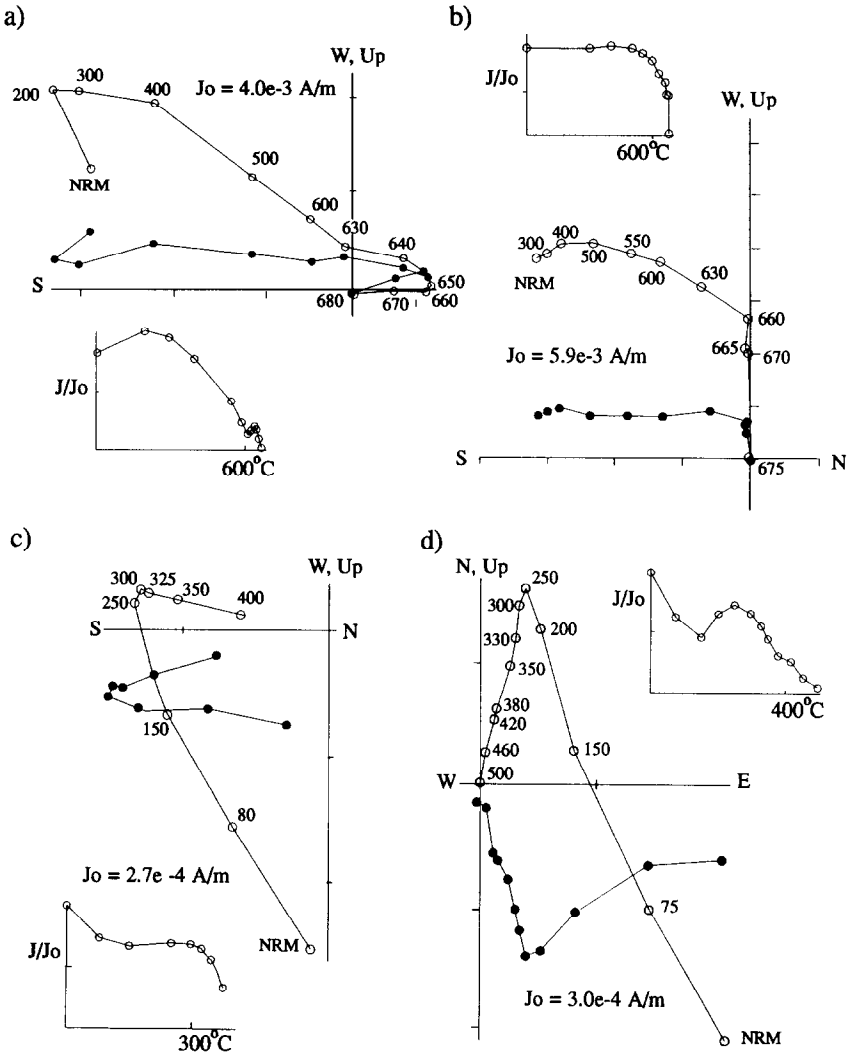


Fig. 2. Representative Zijderveld (1967) diagrams of thermal demagnetization in geographic coordinates. ○ and ● show vector endpoints in the vertical and horizontal plane, respectively. (a) and (b) are Bloomsburg Formation samples showing three components during demagnetization, a present-day field component removed between 25 and 400°C, a secondary (B component) removed between 200 and 660°C and a characteristic magnetization (C component) removed between 660 and 680°C. (c) and (d) are Reedsville shale and Bellefonte dolomite samples showing two components of magnetization, a present-day field component removed between 25 and 300°C and the B component magnetization removed between 250 and 500°C.

b). In the Reedsville and Bellefonte samples two components were isolated: (1) an initial component parallel to the present earth's field with unblocking temperatures less than 300°C; and (2) a higher-temperature B component with unblocking temperatures between 300 and 500°C (Fig. 2c and d).

C component directions were isolated in samples from most of the redbed sites, except for four sites near Watsontown, Pennsylvania (sites 9–12). The magnetic behavior of these samples above demagnetization temperatures of 650°C was unstable, with large increases in bulk susceptibility and strong viscous relaxation of the remanence.

Site mean directions of the Bloomsburg Formation's C component are scattered in geographic coordinates but converge on a northerly and moderately up direction in stratigraphic coordinates (Table 2). Individual fold tests yield both prefolding and synfolding distributions (Table 3; Fig. 3a and b). The fold tests at the Watsontown and Bloomsburg localities yield significant prefolding distributions. The fold tests at the Milton and Cumberland localities yield significant synfolding distributions. The folds at Drehersville, Mount Union and Hancock did not yield significant fold test results, although in all three cases kappa reached a maximum value at or near 100% unfolding. A regional fold test of all sites yields a synfolding pattern but it is not significantly different from the prefolding distri-

Table 2
Paleomagnetic results from the Silurian Bloomsburg Formation, Silurian–Devonian Andreas redbeds, Ordovician Reedsville shale and Ordovicain Bellefonte dolomite

Site	B component						C component					
	N	k	D_g	I_g	D_s	I_s	N	k	D_g	I_g	D_s	I_s
SbA	8	18	164	−3	179	43	4	22	340	−10	8	−56
SbB	6	28	177	19	224	55	9	81	0	11	11	−35
SbC	9	18	172	14	171	−19	6	51	331	−62	10	−28
SbD	10	21	171	17	169	−26	6	44	294	−55	346	−31
SbE	7	101	173	8	174	−17	10	54	21	−43	7	−24
SbF	6	319	177	−22	175	8	10	42	3	3	6	−25
SbG	5	21	166	−17	167	20	7	100	4	−11	358	−52
SbH	5	18	187	19	187	4	1	–	345	−56	349	−41
SbI	3	32	184	11	184	23	1	–	17	−38	17	−50
SbJ	4	32	178	−18	179	7	1	–	20	−36	28	−60
SbK	3	49	174	−2	174	12	0	–	–	–	–	–
SbL	3	24	183	4	189	9	2	–	11	−45	10	−50
SbM	4	132	164	−19	159	36	8	81	5	17	3	−60
SbN	7	80	173	27	176	−27	11	55	211	−86	2	−38
SbO	5	80	162	21	159	−2	8	116	5	−37	349	−24
SbP	5	96	172	−19	170	3	10	110	323	3	324	−25
SbQ	9	93	182	−2	181	1	5	39	312	−13	333	−47
SbR	7	121	188	−1	189	−8	4	56	329	−52	321	−34
SbS	3	45	187	−19	180	7	5	83	341	21	344	−27
SbT	6	46	194	−24	179	5	3	55	327	1	4	−56
SbU	4	186	179	24	176	−19	11	57	82	−57	346	−47
SbV	6	64	194	42	173	−12	3	258	108	−51	330	−45
SDaA	12	22	184	−40	176	−2	14	22	347	−15	358	−55
SDaB	18	36	170	−19	181	−56	19	26	68	−75	358	−50
OrA	10	34	168	12	169	−23						
OrB	8	49	170	−16	172	29						
ObA	8	102	178	−66	166	−18						
ObB	9	102	167	−9	173	−15						

N is the number of samples averaged; k is the Fisher (1953) precision parameter; D_g (D_s) and I_g (I_s) are the declination and inclination of the site mean direction in geographic (stratigraphic) coordinates (°).

bution (Fig. 3c) with a mean direction of 355°/ − 42° ($k = 25$). For sites on the northeastern limb of the salient (sites 1–14, combining sites 8, 9, 10 and 12 into one site) the mean direction is 4°/ − 40° ($k = 31$). For sites on the southwestern limb of the salient (sites 15–22) the mean direction is 338°/ − 38° ($k = 23$).

C component directions from the two new Andreas sites are similar to those reported by Miller and Kent (1988a). Incremental unfolding of these sample directions yields a synfolding pattern (Fig. 3a). However, unlike the results of Miller and Kent (1988a), this synfolding distribution is not statistically different from the prefolding distribution. The mean direction in stratigraphic coordinates is 359°/ − 52° ($k = 25$). The overall mean direction of the Andreas redbeds, combining the two new site mean directions with the site mean directions obtained by Miller and Kent (1988a), is 358°/ − 54° ($k = 126$).

B component directions were isolated in all the redbed sites. In both geographic and stratigraphic coordinates the site mean directions are to the south with moderate to shallow inclinations (Table 2). The mean direction in geographic coordinates for the 24 redbed sites is 176°/0° ($k = 14$). Full tilt correction does not significantly change either the overall mean direction or the dispersion of the directions, 176°/2° ($k = 10$). Incremental unfolding of the directions shows a significant improvement of the dispersion at 45% unfolding, 176°/ − 1° ($k = 23$). The B component directions in the Bellefonte Dolomite and Reedsville are similar. In geographic coordinates the mean direction is 169°/ − 19° ($k = 6$). In stratigraphic coordinates the mean direction is 170°/ − 7° ($k = 12$). Incremental unfolding yields a significant synfolding distribution at 55% bedding tilt, 169°/ − 13° ($k = 16$).

Individual localities show a range of fold test results (Fig. 4). The folds at Drehersville and Cumberland yield significant postfolding distributions. The folds at Bloomsburg, Hancock, Potters Mills, and Andreas yield significant synfolding distributions. The fold at Milton also shows a synfolding pattern, but it is not significant at the

Table 3
Fold locality results for C component directions of the Bloomsburg Formation

Location	N/N_s	Fold test	D_o	I_o	k_p	D_s	I_s	k_s	Θ
Drehersville, PA	25/4	Pre [b]	4	−36	24	4	−36	24	68
Bloomsburg, PA	20/2	Pre	6	−25	49	6	−25	49	79
Watsontown, PA	12/5	Pre	3	−46	49	3	−46	49	99
Milton, PA	19/2	Syn (90%)	3	−43	57	3	−43	44	94
Mount Union, PA	18/2	Pre [b]	337	−24	28	337	−24	28	31
Cumberland, MD	9/2	Syn (70%)	323	−39	51	327	−41	33	27
Hancock, MD	22/4	Pre [b]	345	−44	31	345	−44	31	29
Kent (1988) A, B, C	16/3	Syn (80%)	342	−33	83	346	−39	53	49
Kent (1988) D–I	31/6	Syn (80%)	345	−30	64	345	−28	34	22
Kent (1988) J–L, Q–S	25/6	Syn (90%)	358	−27	83	358	−27	63	76
Kent (1988) O, P	12/2	Pre [c]	2	−37	–	2	−37	–	80
Delaware Water Gap, PA	24/2	Syn (40%)	355	−25	15	358	−21	10	60
Danville, PA [a]	51/7	Syn (70%)	360	−40	74	358	−37	25	80
Round Top, MD [a]	68/11	Pre	346	−28	35	346	−28	35	33

N is the number of samples and N_s is the number of sites for each fold location. Incremental fold test results are prefolding (Pre), postfolding (Post), or synfolding (Syn) as determined from the McFadden and Jones (1981) fold test. For synfolding distributions the number in parantheses shows the increment of unfolding where the directions are best clustered. D_o and I_o are the mean declination and inclination of the location at the optimum bedding correction determined by the fold test (°). D_s and I_s are the declination and inclination in stratigraphic coordinates (°). k is the Fisher (1953) precision parameter. Θ is the azimuth of the fold axis.

[a] Stamatakos and Kodama (1991).

[b] Fold test not significant at 95% confidence level although maximum k is reached at 100% unfolding.

[c] Only two site means used to calculate the location mean.

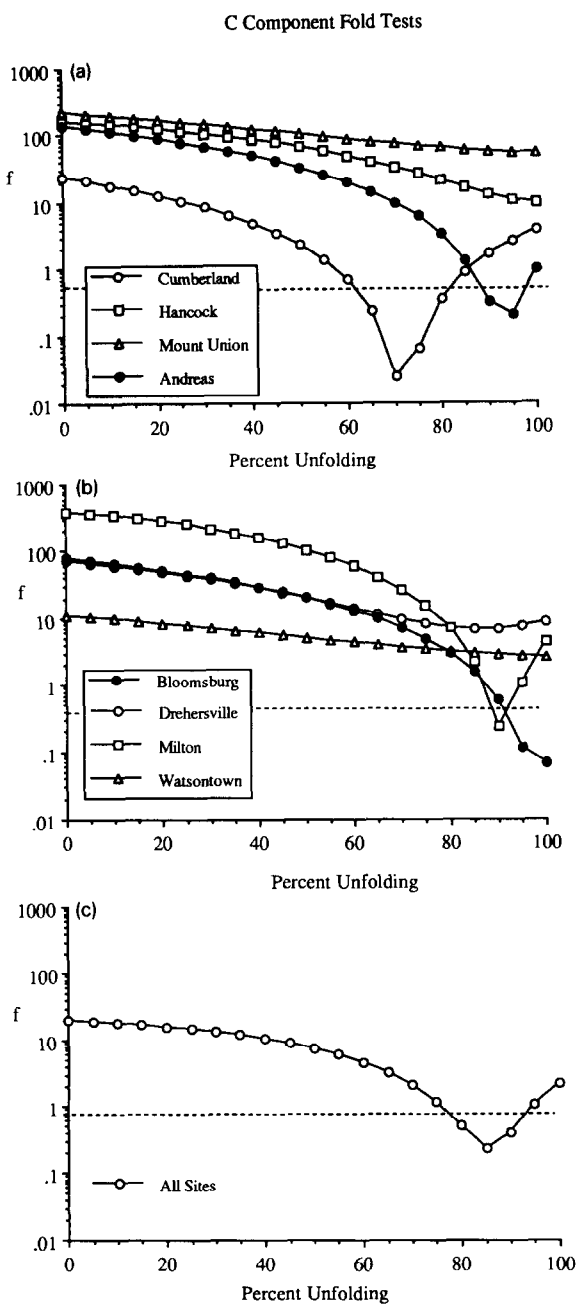


Fig. 3. Incremental fold test for the C Component magnetizations showing the variation of the McFadden and Jones (1981) f parameter as a function of percent unfolding. The f parameter reaches a minimum value when the paleomagnetic vectors from the two fold limbs are least dispersed. Beds were unfolded about line of strike in 5% increments of bedding dip. Dashed lines show the approximate 95% significance level.

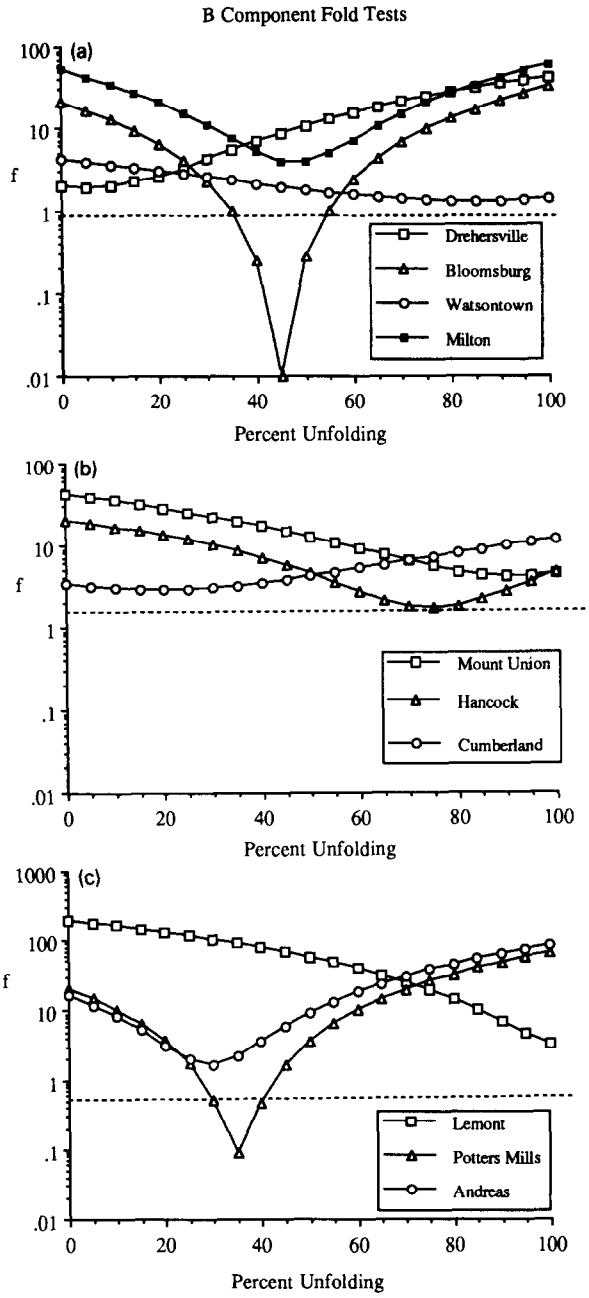


Fig. 4. Incremental fold test results for the B component magnetization as described in Fig. 3.

95% level. The folds at Watsontown, Mount Union and Lemont yield prefolding distributions. Directions for each fold locality, based upon the optimum bedding orientations indicated by the

fold test results, are also to the south with shallow inclinations (Table 4). For localities on the northeastern limb of the salient, the B component mean direction is 172°/1° ($k = 49$). For sites on the southwestern limb of the salient, the mean direction is 173°/1° ($k = 70$).

4.1. Previous results

Previously reported C component results for the Bloomsburg formation were taken from Kent (1988) and from Stamatakos and Kodama (1991a) and are summarized in Table 3. Kent (1988) did not directly sample around individual folds and these site mean directions were grouped into four fold localities based on the site loacation map (fig. 1 of Kent, 1988).

Previously published B component results from sites constrained by local fold tests or from sites without significant bedding corrections are also considered (Table 4, locations 10 to 21). Paleomagnetic and rock magnetic results from the Cambrian Allentown Dolomite and Leithsville Formation show that the characteristic magnetization of these carbonates is carried by authigenic magnetite acquired after Alleghanian fold-

Table 4
Fold locality results for B component directions

Location	N/N_s	Fold test	Θ	D_o	I_o	k	VGP				Rotation			
							Lat	Long	d_m	d_p	P_l	±	P_u	±
Drehersville, PA	33/4	Post	68	171	12	17	43	117	6	3	4	7	0	8
Bloomsburg, PA	13/2	Syn (55%)	79	175	−5	157	51	112	3	2	8	4	5	5
Watsontown, PA	23/6	Pre	99	178	12	21	43	105	7	4	12	7	9	8
Milton, PA	11/2	Syn (45%)	94	171	3	46	46	117	7	3	4	7	1	8
Mount Union, PA	10/2	Pre	31	164	7	67	45	124	7	3	−1	6	−5	7
Cumberland, MD	16/2	Post	27	180	−2	94	51	101	4	2	15	4	12	6
Hancock, MD	19/4	Syn (80%)	29	175	−3	55	51	109	5	2	10	15	7	6
Potters Mills, PA	18/2	Syn (35%)	62	168	−0	41	48	120	5	3	2	6	−1	7
Lemont, PA	17/2	Pre	57	170	−17	89	56	123	4	2	3	5	−1	6
Water Gap, PA[a]	22/2	Post	60	175	−15	22	56	114	7	4	7	7	4	8
Danville, PA[a]	49/8	Syn (40%)	80	178	0	55	49	107	8	4	11	5	7	7
Andreas,PA[b]	59/4	Syn (40%)	63	174	−25	88	62	178	11	6	6	11	3	12
Lavelle, PA[c]	76/10	Syn (85%)	75	176	3	217	48	111	3	2	9	4	6	5
Allentown, PA[d]	25/4	Post	60	173	5	81	47	114	8	4	6	8	3	9
Easton, PA[d]	15/3	Post	65	175	−7	95	53	113	8	5	7	9	4	9
Alexandria, Pa R[e]	24/2	Pre	35	180	−6	–	52	103	7	–	14	7	11	8
Alexandria, Pa G[e]	11/2	Pre	35	179	7	–	46	103	14	–	14	14	11	15
Anthracite Basin[f]	24/4	Flat Lying	50	179	−3	33	51	106	16	8	12	17	9	17
Broad Top, Pa[g]	16/3	Flat Lying	34	160	−5	132	49	132	11	5	−5	11	−8	12
Round Top, MD[a]	77/9	Syn (40%)	33	165	−5	24	51	126	11	6	−1	11	−4	12
Catskill, NY[h]	43/9	Flat Lying	31	172	1	116	47	117	5	2	4	5	6	3

Symbols are the same as in Table 3. VGP Lat and Long are the latitude (°N) and longitude (°E) of the virtual geomagnetic pole; d_p and d_m are the axes of the 95% confidence ellipse around the VGP. P_l and P_u are the rotation (°) of the location needed to bring the location VGP into agreement with the Early and Late Permian reference poles for North America (Van der Voo, 1990), positive is clockwise. ± are the confidence limits of these rotations (°) calculated from the 95% confidence regions around the VGP's and reference poles following the method suggested by Demarest (1983).

[a] Stamatakos and Kodama (1991a).
[b] Miller and Kent (1988a) combined with results in Table 1.
[c] Stamatakos and Kodama, (1991b).
[d] Stead and Kodama (1984).
[e] Perry (1991), only pole positions, A_{95} and fold test results reported.
[f] Kent and Opdyke (1985).
[g] Kent (1988).
[h] Kent and Opdyke (1978).

ing in the Permian (Stead and Kodama, 1984). Site mean directions for both formations are scattered in stratigraphic coordinates but cluster around a south and shallow direction in geographic coordinates. Local fold tests of the Leithsville samples yield a significant postfolding distribution. Incremental unfolding of Allentown Dolomite sample directions from a fold in Bethlehem, Pennsylvania shows maximum kappa at 15% unfolding (Kodama, 1988). This synfolding distribution is significant using the McElhinny (1964) criteria. However, based on the McFadden and Jones (1981) fold test, this synfolding distribution is not significantly different from the postfolding magnetization. Thus, for these rocks, the postfolding site mean directions were combined to produce locality mean directions for the cities of Allentown and Easton, Pennsylvania (Table 4).

Perry (1991) examined the magnetization of a single 30-cm-thick bed of Clinton Ironstone on both limbs of a syncline near Alexandria, Pennsylvania. The bed contains gray, chamosite-rich margins which grade into a red hematite-rich interior. Thermal demagnetization of the gray samples reveals two magnetic components: an initial component aligned along the present earth's field direction and a characteristic component with unblocking temperatures between 300 and 580°C. Two magnetic components were also isolated in the red samples: an initial viscous component parallel to the present earth's field direction and a B component with distributed unblocking temperatures between 300 and 620°C. Above 620°C the samples showed strong viscous relaxation of the remanence directions which obscured further demagnetization of the characteristic magnetization. Both red and gray samples yield similar directions, which cluster about a south and shallow direction after correction for bedding tilt. For both red or gray sample directions, incremental fold tests yield significant prefolding distributions.

Additional B component results were also obtained from Bloomsburg Formation directions from three folds in Pennsylvania and Maryland (Stamatakos and Kodama, 1991a), Catskill Formation results from the flat-lying Devonian section of the Appalachian Plateau in New York (Kent and Opdyke, 1978), and Mauch Chunk Formation results from the Frackville anticline in central Pennsylvania (Stamatakos and Kodama, 1991b) and from flat-lying sites on the Broad Top Basin in southern Pennsylvania (Kent and Opdyke, 1985). Mean directions for each fold locality were calculated using the optimum bedding attitude indicated by the fold test results. Mean directions for the flat-lying localities at Broad top and the Catskill plateau were calculated by combining the respective site mean directions. Adding these B component directions to the new results presented above yields an overall mean direction for the northeastern limb of the salient of 174°/ − 3° ($k = 54$) and an overall mean direction for the southwestern limb of 172°/ − 1° ($k = 62$).

5. Discussion

Three factors must be considered in order to evaluate these paleomagnetic results in terms of the structural evolution of the Pennsylvania salient: (1) the fold test results, which constrain the relative timing of the magnetizations to folding across the region; (2) the ages of the magnetic components, which constrain the age of the deformation around the salient; and (3) the distributions of the magnetic declinations, which are used to determine the amount and sense of rotations around the salient. In the discussion that follows, these three factors are considered separately for the C and B components. From these results a tectonic interpretation of the structural evolution of the salient is presented.

5.1. C component results

Fold test results

Analysis of the relationship between penetrative strain and remanence in several Bloomsburg and Mauch Chunk folds across the Valley and Ridge indicate that the larger hematite grains, those that appear to carry the C component directions, actively reorient relative to the rock matrix during deformation (Stamatakos and Kodama, 1991a,b). For penetrative strains that accu-

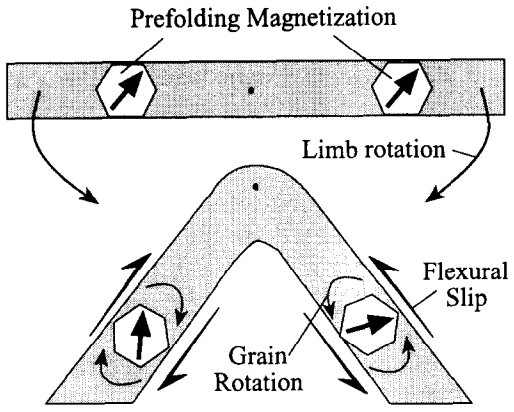


Fig. 5. Schematic model showing that internal rotations associated with flexural slip and flexural flow folding is opposite to the external rotation of the fold limbs.

mulate during the growth of folds, this reorientation of remanence can significantly alter the fold-test results (Facer, 1983; Van der Pluijm, 1987; Kodama, 1988). For example, on fold limbs that develop with a significant component of flexural slip and flexural flow, the sense of shear within the beds is opposite to the external rotation of the beds about the fold axis (Fig. 5). Reorientation of the C component directions by both fold-limb rotation and internal bedding-parallel shear in such folds results in in-situ remanence directions that are "under-rotated" relative to the dip of bedding. Full bedding correction overcorrects these directions such that the original prefolding magnetizations now mimic synfolding distributions.

Five of the C component fold tests in this study suggest that the C component magnetization is prefolding (Table 3). The other two folds, at Milton and Cumberland, yield synfolding distributions at between 70 and 90% unfolding. However, because of the nature of this synfolding distribution and the similarity of these folds to those analyzed by Stamatakos and Kodama (1991a,b), both in terms of fold geometry and strain patterns observed at the outcrop, these C component synfolding distribution are also considered to be strain-modified prefolding magnetizations.

Age of magnetization

The prefolding age of the Bloomsburg Formation's C component magnetization only indicates that this magnetization is pre-Alleghanian (Early Pennsylvania to Late Permian). Unfortunately, there are no available field tests (conglomerate or baked contact tests) that can directly constrain the absolute age of the Bloomsburg C component magnetization. However, several observations suggest that this magnetization is in fact primary: (1) Irving and Opdyke (1965) reported 20 mm hematite in thin sections of Bloomsburg Formation samples which they interpreted as detrital specular hematite. According to the Néel equation, this is the grain-size fraction of hematite that would have high unblocking temperatures similar to the unblocking temperatures of the C component magnetization. (2) Stamatakos and Kodama (1991a) noted antipodal C component directions from the Bloomsburg section exposed at Round Top Maryland. These antipodal directions are stratigraphically controlled. (3) The pole positions for Bloomsburg Formation sites with relatively low strain values fall on the Late Silurian part of the North American Polar wander path for North America (Stamatakos and Kodama, 1991a).

Rotations

The mean direction of Bloomsburg Formation sites northeast of the salient have similar inclinations but significantly different declinations than those southwest of the salient, in agreement with the results of Kent (1988). The difference in inclinations is $2 \pm 11°$. The difference in declination is $25 \pm 13°$. Combining these results with those of Kent (1988) yields a mean direction for the northeastern limb of $1°/-34°$ ($k = 32$) and a mean direction for the southwestern limb of $343°/-34°$ ($k = 30$). The resulting difference in inclination is $0 \pm 7°$ and the resulting difference in declination is $18 \pm 8°$. Thus within the 95% confidence limits, the C component inclinations on the two limbs of the salient are statistically indistinguishable, but the C components declinations are significantly different. These results indicate up to 26° of relative clockwise rotation of the northeastern limb of the salient or counter-

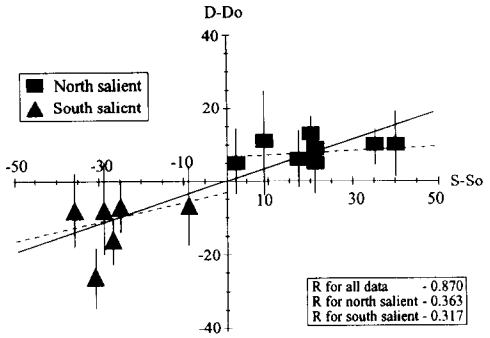


Fig. 6. Declination (D) of the Bloomsburg Formation C component directions minus the average declination (D_o) plotted as a function of fold-axis trend (S) minus average trend (S_o). Error bars are the α_{95} of the fold-location mean directions. Solid line is the best-fit linear regression for all data points. Dashed lines are the best-fit linear regressions for the individual limbs of the salient. R is the correlation coefficient.

clockwise rotation of the southwestern limb. Similar results were reported by Kent (1988) for both the Bloomsburg and Mauch Chunk formations.

The relationship between the C component declinations and the orientation of fold axes around the salient is shown in Fig. 6. The overall correlation between declination and structural trend is significant and reinforces the conclusion that there was up to 27° of relative rotation between the two limbs of the salient sometime during the development of the orogen. However, closer examination of these data show that within each limb of the salient the declinations do not directly correlate with the trend of the fold axes. This lack of a direct correlation between the fold-axis trend and declination indicates that the simple model of an initially more linear fold and thrust belt that was bent into its present curved form by post-orogenic (oroclinal) bending is incorrect. The structural development of the salient appears to involve a much more complex history of rotations and folding.

Two other theories previously proposed to explain the difference in C component declinations between the two limbs of the salient are rapid APW and strain modified declinations. However, careful analysis of the available data show that these explanations are less likely alternatives.

Kent (1988) noted that the differences in declination around the salient could be the result of rapid APW if the magnetizations were acquired at different times on the two limbs of the salient. For example, acquisition of the characteristic magnetization in the Bloomsburg Formation for sites on the northeastern and southwestern limbs of the salient would have been separated in time by approximately 25° of APW. Direct testing of this hypothesis requires a well resolved and independently determined APW path for North America. At present, there are too few Paleozoic reference poles from the North America craton to meet these criterion. However, as noted by Kent (1988), this hypothesis seems unlikely because it requires that a very specific sequence of remanence acquisitions and APW was repeated in the Silurian, Devonian and Carboniferous.

Stamatakos and Kodama (1991a) suggested that the difference in C component declinations on the two limbs of the salient may have resulted from grain reorientation during deformation. In this model, the unstained directions on both limbs of the salient would be similar (steeply up toward the north). Because shortening directions vary around the salient, greater amounts of penetrative strains from shortening would result in progressively divergent paleomagnetic declinations around the salient. On the northeastern limb, shortening directions are south to north and thus, greater penetrative strains would flatten the inclinations but keep the declinations pointing north. On the southwestern limb of the salient, shortening directions are southeast to northwest, and thus greater penetrative strains would result in shallower inclinations and more northwesterly declinations. This pattern was initially observed at two Bloomsburg Formation sites. C component directions from sites at Round Top in Maryland and at Montour Ridge in Pennsylvania with low strain have similar north and moderately up directions (Fig. 7a). At Montour Ridge the directions become progressively shallower as strain values increase but their declinations remain toward the north. In contrast, directions from the Round Top show both shallower inclinations and more northwesterly declinations as strain values increase.

This model of grain rotation suggests that there is a direct relationship between inclination and declination on the southwestern limb of the salient. As the rocks are internally strained, their inclinations become shallower and their declinations deflect toward the northwest. All the Bloomsburg Formation data from the southwestern limb of the salient can be used to test this model. As shown in Fig. 7b, the explicit relationship between inclination and declination does not exist. Thus, although strain may have played an important role in altering the remanent directions of these rocks, especially in influencing the outcome of the C component fold test, this simple model of grain reorientation does not explain the observed differences in declination between the two limbs of the salient.

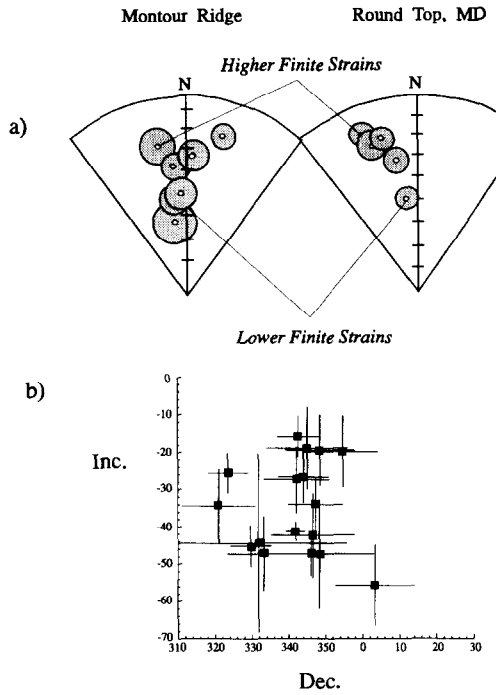


Fig. 7. (a) Equal area projection of the variation in Bloomsburg Formation C component directions for the Round Top fold in Maryland and the Montour Ridge fold in central Pennsylvania (Stamatakos and Kodama, 1991a). Sites with relatively low finite strain values share a common steeply up and northern direction. As finite strain values increase, the Montour ridge directions move to shallower inclinations but stay oriented toward the north. In contrast, the Round Top directions move toward shallower inclination and more northwesterly declinations. (b) Declination of the C component directions for sites on the southwestern limb of the salient plotted as a function of inclination.

5.2. B component results

Fold test results

Unlike the C component, it appears that the smaller hematite grains, those that carry the B component magnetization, are unaffected by the penetrative strains. Plots of inclination or declination against finite strains show the B component directions do not correlate with finite strain (Fig. 8). Several factors or combinations of factors may explain these results: (1) Assuming a rigid particle model (Jeffrey, 1923), the small grain size of the B component prohibits it from rotating as a rigid clast in the more viscous matrix. (2) The B component may have been acquired after much of the penetrative strains accumulated in these rocks. Recent structural analyses indicate that prefolding layer-parallel shortening and layer-parallel shear are important components of the progressive deformation of the central Appalachians (Gray, 1991). (3) The mean orientation of B component grains is sub parallel to the shear plane (bedding). This orientation of grains is least affected by bedding-parallel shear. Moreover, strain values may be too small to rotate these grains out of the shear plane. Therefore, unlike the C component results, B component fold tests appear to reliably indicate the timing of remanence acquisition relative to folding across the region.

The effects of strain on magnetite-bearing rocks is less clear. Kodama (1988) argued that penetrative strains in the Allentown Dolomite were too small to affect the magnetization of these rocks. In contrast, Housen and Van der Pluijm (1991) concluded that remagnetization of the Martinsburg Formation at Lehigh Gap, Pennsylvania is controlled by dissolution and neocrystallization of magnetite and subsequent grain reorientation related to the formation and development of slaty cleavage. Growth of new magnetite during deformation yields directions representative of the earth's magnetic field at the time of remanence acquisition. Grain rotation can yield

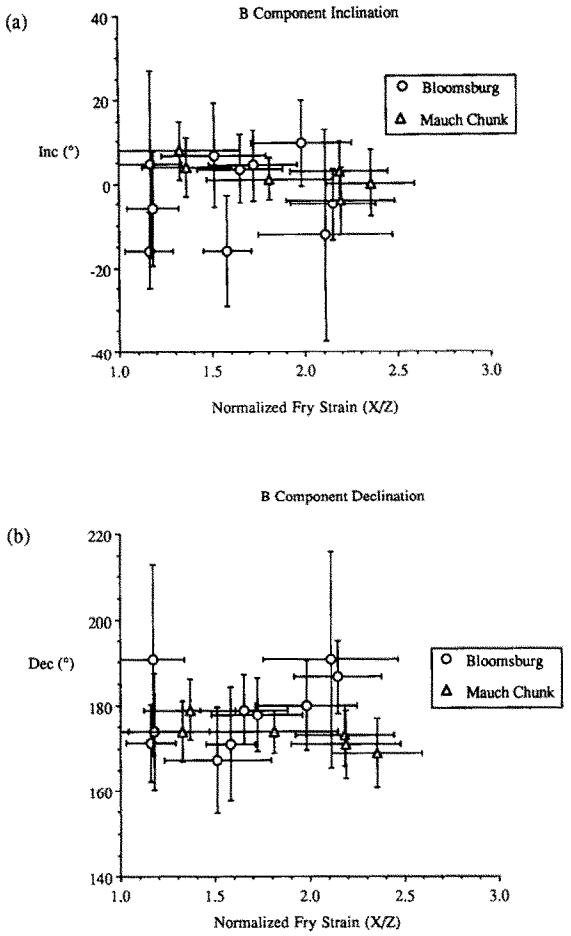


Fig. 8. Inclination and declination of the B component magnetizations as a function of finite strain for sites at Delaware Water Gap, Montour Ridge, and the Frackville Anticline (Stamatakos and Kodama, 1991a,b).

an infinite number of resultant magnetic directions depending upon such factors as the initial orientation of the magnetic particles, the types magnetic minerals, and the conditions of the imposed deformation. Unfortunately, because the rocks at Lehigh Gap have a uniform dip, these effects can not be related to folding.

The rocks exposed at Bellefonte and Lemont sampled for this study do not show the penetrative slaty cleavage observed at Lehigh Gap. Rather, these rocks appear to have internal strains similar to other Valley and Ridge carbonates, like the Allentown Dolomite. Hence, it is assumed that the fold tests and mean directions for these folds are reliable.

Age of magnetization

Comparison of the B component paleomagnetic poles with the reference polar wander path for North America indicates that the B component magnetizations were acquired in the Late Carboniferous or Early Permian. These results are consistent with the conclusion that many of the rocks within the Appalachian basin and the adjoining North American craton were remagnetized during the Alleghanian orogeny (McCabe and Elmore, 1989). Miller and Kent (1988b) showed that the age of the remagnetization is older in the southern Appalachians than the central Appalachians. They argued that this age difference represents south to north progression in the age of the Alleghanian orogeny. However, more recent analysis of the timing of remagnetization relative to deformation suggests that although the remagnetization may be younger in the central Appalachians than the southern Appalachians the trend is not progressive (Stamatakos and Hirt, in prep.). The age of the remagnetization within the central Appalachian domain is consistently Early Permian in age.

Rotations

Unlike the C component, B component directions from the two limbs of the salient are similar. Comparison of the mean B component directions yield a difference in inclination of $2 \pm 7°$ and a difference in declination of $2 \pm 8°$. Thus, the B component directions, in inclination as well as declination are statistically similar on both limbs of the salient. Comparison of the pole positions, calculated from these B component directions (Table 4), with the Early or Late Permian reference poles for North America (Van der Voo, 1990) also shows that there has been no significant rotation around the salient since the acquisition of this late Paleozoic magnetization (Figs. 9 and 10). The B component pole positions are not significantly rotated from the reference polar wander path for North America (Fig. 10c) even if the age of remagnetization is considered older in the Maryland sites, as suggested by Miller and

Kent (1988b). Because these remagnetized directions are generally synchronous with folding and in some cases predate folding (Table 4), these results also indicate that the curvature of the fold axes around the salient did not result from bending of initially more linear folds and thrusts.

5.3. Tectonic interpretation

Paleomagnetic analysis of the B and C component results indicates that the structural evolution of the Pennsylvania salient is more complex than either a simple rotation or nonrotational model would suggest. The B component results do not indicate significant rotation of the salient after the rocks were remagnetized in the Late Permian, even though this remagnetization took place before the folds and thrusts fully developed in the central Appalachian foreland. In contrast, the C component results suggest rotations that are consistent with but not directly related to the curvature of the major folds and thrusts around the salient. One possibility is that the rotations indicated by the C component results are pre-tectonic or associated with deformational events that occurred early in the structural evolution of the

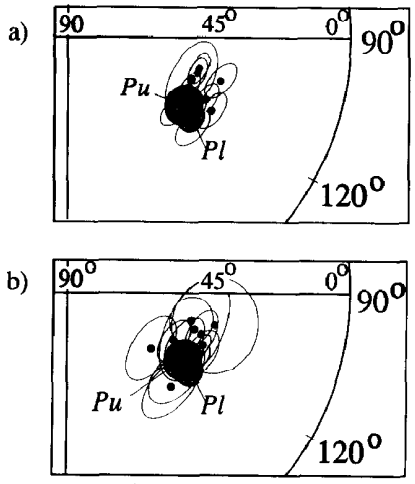


Fig. 9. Equal angle projection of the virtual geomagnetic pole positions for the B component results plotted against the Late and Early Permian reference poles for North America (Van der Voo, 1990). New results are shown in (a), previously reported results shown in (b).

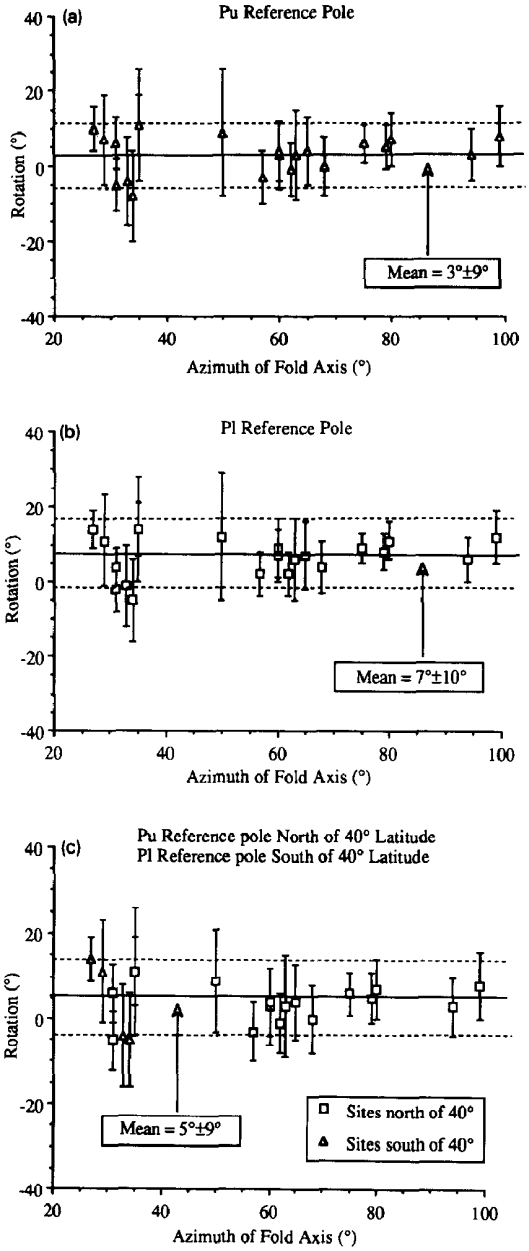


Fig. 10. Angular difference (rotated about the site) between the B component poles and the Late (a) or Early (b) Permian reference poles for North America (Van der Voo, 1990) plotted as a function of fold-axis trend. In (c), localities south of 40° latitude were compared with the Early Permian pole while localities north of 40° latitude were compared against the Late Permian reference pole.

salient, prior to folding and faulting of the foreland. The later folds and thrusts simply mimic the geometry of this earlier rotational phase without further bending of the orogen. The stratigraphic and paleomagnetic evidence place this rotational phase sometime between the earliest Pennsylvanian and the Late Permian. The Mauch Chunk Formation was deposited in Late Mississippian or possibly earliest Pennsylvanian times (Wood et al., 1969). Foreland folding and thrusting, constrained by the age of the synfolding remagnetization, occurred in Late Permian times (Miller and Kent, 1988b). This conclusion is consistent with both the two-stage Alleghanian orogeny envisioned by Geiser and Engelder (1983) and with a multi-sequence Alleghanian orogeny in which much of the deformation occurred prior to folding across the fold and thrust belt (Nickelsen, 1979; Gray and Mitra, 1993).

6. Conclusions

Detailed analysis of both primary magnetizations and the late Paleozoic remagnetization in the central Appalachian foreland shows that the curvature of the major folds and faults around the Pennsylvania salient is an inherited feature of the orogen and not the result of subsequent bending of initially more linear folds and thrusts. Synfolding and even prefolding remagnetized directions from sites on both limbs of the Pennsylvania salient are similar and not significantly rotated from either the Early Permian or Late Permian reference poles for North America. In contrast, the mean declination of the Bloomsburg Formation's characteristic magnetization on the northeastern limb of the salient is significantly different from the mean declination of the southwestern limb. These differences in declination are in agreement with previous C component results for the Bloomsburg and Mauch Chunk formations (Kent, 1988). However, declination of the characteristic directions from the Bloomsburg Formation do not directly correlate with trends of first-order folds around the salient. This difference in declination does not appear to be an artifact of internal strains or rapid APW. A more favorable explanation is that these differences in declination resulted from pre-orogenic rotations or rotations associated with deformational events that occurred early in the structural development of the region, prior to remagnetization and folding of the Appalachian foreland.

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